SCHEMATIC STATUS REPORT

AO-Horizon Power Sheet Status Report

State of the saved power_supplies.kicad_sch, findings and next steps

Phase	Spec	Sheet	ERC
DRAFT	**rev 0.17**	**power_supplies (28 parts)**	**not run**

Document	AOH-RPT-001
Revision	A
Date	Oct 28, 2026
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CERN-OHL-S-2.0

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Natchanon Thuentham · Oct 28, 2026 · Doc. No. AOH-RPT-001 · Rev. A

The power sheet (`power_supplies.kicad_sch`) now holds 28 parts and is mostly wired. The TPS16412DRC eFuse connects +VIN to +5V, the TPS62142RGTR buck makes +3V3 and the TPS73533DRBR LDO makes +3V3A for the two IMUs. Two items need work before the sheet can be reviewed: the eFuse EN/~SHDN pin has no connection, and six eFuse part values are still ?. One part-number code (LCSC C22548) was copied onto resistors of other values. All parts are unannotated and ERC has not been run. This report records what is drawn. It does not approve the design.

1. Scope and status

This report covers one sheet of the AO-Horizon flight controller: the power input and the 3.3 V rails. It is for the AeroOptix team, so everyone sees the same state before the next design step.

Table 1. Status and sources

Item	Value
Project phase	DRAFT (declared in the root README)
Specification	`AO-HORIZON_SPEC_EN.md` rev 0.17, 2026-10-28
Checklist	`AO-HORIZON_EXECUTION_CHECKLIST_EN.md` rev 0.16
Design baseline	Not established
File read	Saved `power_supplies.kicad_sch`, uncommitted changes after commit `e15be06`
Tools	`sch-inspect` pin and net queries, `progress-update` preflight
ERC / DRC	Not run

> **Not a verification record.** Every value in this report keeps the evidence label it has in the specification. ERC and DRC have not run, and a later ERC pass would not prove the circuit correct either.

The other sheets (root, `top_hierarchical`, `block_diagram`, `revision_history`) hold no parts. The PCB has no footprints and no board outline.

2. Power tree

The sheet takes 5 V from the PM02D power module and makes two 3.3 V rails. Blue blocks are decided parts; gray blocks are drawn but not decided:

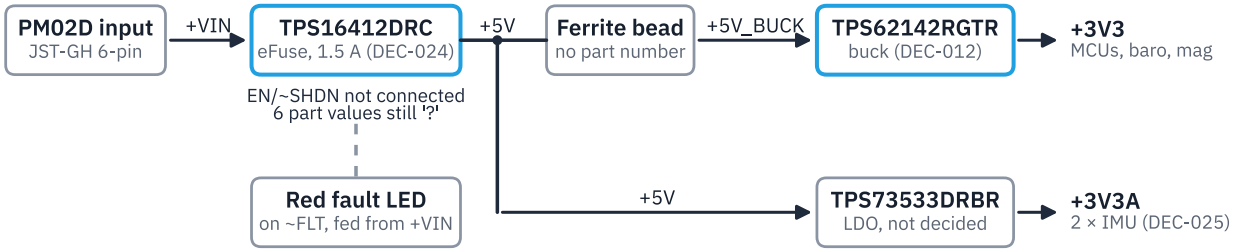


Figure 1. Power tree as drawn on the saved sheet. Blue: decided part (DEC). Gray: drawn, not decided.

- +VIN is the PM02D 5 V input. The eFuse limits the board current to 1.5 A (DEC-024).
- +5V feeds the buck through a ferrite bead (+5V_BUCK) and feeds the LDO directly.
- +3V3 supplies the MCUs, barometers and magnetometer. +3V3A supplies only the two ICM-45686 IMUs, on VDD and VDDIO (DEC-025).
- USB VBUS also powers the board on the bench. How it is joined to +5V is still open (OPEN-006).

3. Decisions in force

Three decisions from the specification apply to this sheet:

Table 2. Decisions that apply to the power sheet

ID	Decision	Still open
DEC-012	TI TPS62142RGTR buck, 5 V to 3.3 V	3.3 V load current, inductor and output-capacitor derating
DEC-024	TI TPS16412DRC eFuse between the PM02D +VIN and +5V, current limit 1.5 A	1.8 A rating, ILIM resistor, reverse current with USB, 5 V budget (OPEN-012)
DEC-025	+3V3A feeds only the 2 IMUs, VDD and VDDIO	LDO part, load, SPI level match with the H743, power-up order (OPEN-011)

Every other part on the sheet is drawn, not decided. That includes the TPS73533DRBR LDO, the connector, the ferrite bead, the LEDs and all passives.

4. Power sheet as drawn

The power net or label on each pin comes from `sch-inspect`. Which parts share an unlabeled net (for example the eFuse ILIM resistor) comes from a trace of the wire geometry, because unannotated parts block the project-level net query. Those groupings are `NOT_VERIFIED` until the parts are annotated and ERC runs:

Table 3. Blocks on the saved sheet, 2026-10-28

Block	As drawn
Power input	6-pin JST-GH (`SM06B-GHS-TB`): pins 1-2 `+VIN`, 3 `PM02_SCL`, 4 `PM02_SDA`, 5-6 and MP `GND`. The I2C labels reach only the connector. The planned CLIK-Mate connector is not drawn yet.
`+VIN` to +5V	TPS16412DRC: IN and VCC to `+VIN`, OUT to +5V, GND to `GND`. OVP divider, ILIM and IOCP/IMON resistors, IDLY and dV/dt capacitors, all value `?`. Red fault LED on ~FLT, fed from `+VIN` through 2 kΩ. EN/~SHDN not connected. 2 × 10 µF on +5V.
3.3 V buck	TPS62142RGTR: VIN and EN on `+5V_BUCK` (ferrite bead from +5V, 2 × 10 µF). 2.2 µH inductor to +3V3, 3 × 10 µF output. FB, DEF, FSW (2.5 MHz) and PG to `GND`, SS/TR 3.3 nF. No 0.1 µF AVIN capacitor.
+3V3A LDO	TPS73533DRBR: IN and EN on +5V with 10 µF. OUT on +3V3A with 10 µF and 100 nF. NR/FB 100 nF. NC pins 2, 6 and 7 have no no-connect marker.
Indicators	Green LED with 1 kΩ on +3V3 (not driven by PG). Red fault LED on the eFuse (above).
Power flags	One `PWR_FLAG`, on `+VIN`.

Since the last table in the specification (rev 0.7, 2026-10-19), the DNP DF2S6.2CT ESD diode was removed, the eFuse was wired, and the ferrite bead and +5V_BUCK node were added. The LDO output changed from 2 × 1 µF to 10 µF and 100 nF.

5. Findings

Each finding has a severity and an evidence label. Severity rates one finding, not the whole sheet:

Table 4. Findings on the saved sheet

Severity	Item	Label	Finding
MAJOR	eFuse EN/~SHDN (pin 5)	VERIFIED	No wire, label or no-connect marker. Whether the pin may float is not checked against TI data, so the eFuse may not turn on.
MAJOR	LCSC C22548	VERIFIED	The 1 kΩ part number (RC0603FR-071KL) is also on the 2 kΩ resistor and the 4 resistors with value ?. A BOM built now orders wrong values.
MINOR	eFuse passives	VERIFIED	6 values are ?; the IDLY and dV/dt capacitors have no footprint.
MINOR	Ferrite bead	VERIFIED	No footprint and no part number.
MINOR	LDO NR capacitor	VERIFIED	100 nF, above TI's 0.01 µF limit for the TPS735 (TI SBVS087M, OPEN-011).
MINOR	TPS62142 AVIN	VERIFIED	TI recommends 0.1 µF from AVIN to AGND close to the IC; none is drawn (TI SLVSAJ0F).
MINOR	Red LED fields	VERIFIED	MFR field says LTSL-C191TGKT, the green LED.
MINOR	All parts	VERIFIED	28 parts unannotated.
INFO	Unlabeled nets	NOT_VERIFIED	Groupings come from wire geometry, not a netlist.

VERIFIED here means the statement matches the saved file (or the TI source named). It does not mean the circuit was verified.

6. Open items

These open items from the specification touch this sheet:

Table 5. Open items for the power sheet

ID	Open item	Owner
OPEN-006	USB VBUS and PM02D 5 V joining, VBUS current limit, USB-only load	Hardware
OPEN-011	+3V3A LDO part, load, NR capacitor, IMU SPI level match and power-up order	Human owner
OPEN-012	eFuse EN/UVLO connection, ILIM, OVP, IMON, IDLY and dV/dt values from TI data, reverse current with USB	Hardware
OPEN-023	PM02D connector: change JST-GH to a 6-pin 2.00 mm CLIK-Mate to match its cable	Hardware

OPEN-017 is outside this sheet but is the most important safety question for the team. The 4 × Skywalker 80A V2 ESCs look PWM-only, so the motors may keep running after an H743 hang. It is asked in integration question D13.

7. Next steps

In this order, from checklist tasks P2-011 and P2-017:

1. Connect the TPS16412DRC EN/~SHDN pin.
2. Set the eFuse values from TI data: ILIM for 1.5 A, IOCP/IMON, the OVP divider, IDLY and dV/dt.
3. Pick the +3V3A LDO (OPEN-011). If TPS73533DRBR stays, change its NR capacitor to 0.01 µF or less.
4. Add the 0.1 µF AVIN capacitor at the TPS62142.
5. Give the ferrite bead a part number and footprint.
6. Fix the copied MPN, LCSC and MFR fields.
7. Change the PM02D connector to CLIK-Mate (OPEN-023).
8. Annotate all parts, run ERC on the sheet, then hold a schematic review.

From the team: the answer to D13 (ESC signal loss) and the parts table in integration questions rev 0.4.

8. Evidence and limits

- Read from the saved `.kicad_sch` file only. The KiCad editor was not open.
- `sch-inspect` gave pin positions, local wires, labels and power symbols. It could not run the project-level net query, because the parts are unannotated.
- No datasheet was read for this report. The TI limits quoted in sections 5 and 6 come from earlier spec records (rev 0.7), checked then through NotebookLM against TI SLVSAJ0F Rev. F and SBVS087M.
- ERC, DRC, simulation and bench measurement: none done.